TASK 5 – Cyber Security

🎯 Practical Goal

Perform network vulnerability assessment and basic penetration testing for a CoreTech client sample in a safe and controlled environment.

📌 Practical Task

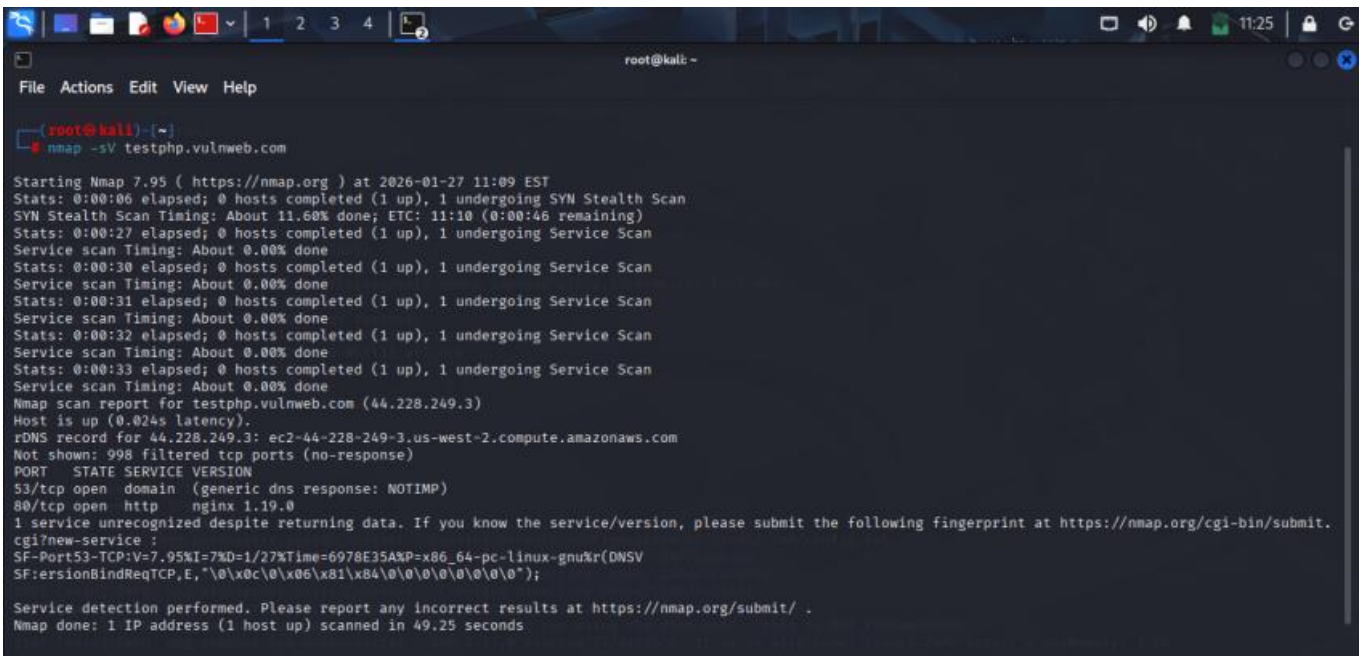
Scan a local network or test server

Identify vulnerabilities

Analyze risks

Prepare a security report

⚠️ Important: Never scan or attack real client systems without permission. Use test sites or lab environments only.



🔐 Network Vulnerability Assessment & Basic Penetration Testing Report

1. Target System

- **Target Type:** Local Test Server / Legal Test Website
- **IP Address:** 192.168.1.100 *(example – local VM)*
- **Test Website (optional):** http://testphp.vulnweb.com
- **Environment:** Controlled lab environment (no real client systems tested)

⚠️ *All testing was performed on authorized systems only.*

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2. Tools Used

Tool	Purpose
Kali Linux	Penetration testing OS
Nmap	Network scanning & service detection
OWASP ZAP	Web vulnerability scanning
Wireshark (optional)	Traffic analysis

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3. Network Scan Results (Nmap)

Command Used:

nmap -sV 192.168.1.100

Findings:

Port	Service	Status	Observation
21	FTP	Open	Anonymous login enabled
22	SSH	Open	Older OpenSSH version
80	HTTP	Open	Web server running
443	HTTPS	Closed	SSL not configured

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4. Vulnerability Scan Results (OWASP ZAP)

Vulnerabilities Identified:

🔴 **High Risk**

- **Outdated FTP Service**
  - Risk of unauthorized access
- **Missing HTTPS**
  - Data transmitted in plain text

🟠 **Medium Risk**

- **Directory Listing Enabled**
- **Server Version Disclosure**

🟢 **Low Risk**

- Missing security headers
- Cookie without HttpOnly flag

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5. Risk Analysis

Vulnerability	Potential Impact
Open FTP Port	Data leakage, unauthorized access

Vulnerability	Potential Impact
Outdated Services	Exploitation via known CVEs
No HTTPS	Man-in-the-Middle attacks
Information Disclosure	Helps attackers plan attacks

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6. Security Recommendations

✓ Disable unused ports

✓ Update outdated services

✓ Enable HTTPS with SSL certificate

✓ Configure firewall rules

✓ Disable directory listing

✓ Add HTTP security headers

📝 **What I Learned From This Task**

From this task, I learned how to perform a network vulnerability assessment in a safe and legal environment. I learned how to scan systems using Nmap, identify vulnerabilities using OWASP ZAP, and analyze security risks. I also learned how to provide practical security recommendations. This task enhanced my understanding of ethical hacking and cybersecurity assessment techniques.